

Solution Design

Ledgerline Cloud Infrastructure Improvement — Approved Design

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1. Purpose and Scope

This Solution Design sets out the improvements to the cloud infrastructure of the YAT Accounting System (Ledgerline). Following YAT's India-campus partnership, the infrastructure must be confirmed stable, reliable, fit for purpose, and compliant with the applicable Indian regulatory requirements, and improved across security, reliability, scalability and cost. The design improves the existing single-Availability-Zone deployment in Sydney; it is the basis for the implementation (the infrastructure is provisioned as code) and the as-deployed verification.

In scope: the cloud infrastructure across four components — network, compute, database, and storage — and a light India residency slice for regulatory logs and financial records. Out of scope: the Ledgerline application and its financial data, which are preserved unchanged (IR-4); the legal interpretation of the Indian obligations (owned by YAT Compliance — treated here as fixed inputs); and payroll (outsourced, not on this system). Improvements are proportionate to an internal, business-hours finance system (IR-2) — sound engineering, not gold-plating — and each is justified on a cost-versus-benefit basis (IR-6).

2. Design Inputs and Requirements

2.1 Inputs

This design is built from the Improvement Requirements (IR-1 to IR-7), the Indian Regulatory Requirements, the Accounting System Cloud Architecture Baseline Design, the Infrastructure Specifications, and the Operational Costing. Ledgerline is an internal, staff-only, business-hours workload (~60 users, ~15-25 concurrent, peaks of ~45-55 at month-end and EOFY, effectively idle overnight and at weekends), reached over the Site-to-Site VPN — not public-facing. The application runs on Amazon Linux with an Amazon RDS for PostgreSQL database; financial records carry a 7-year retention obligation.

2.2 Requirements the design must meet

Ref	Requirement	Source
IR-1	Stable and reliable in business-hours operation; at least preserve the current service level	Improvement Requirements
IR-2	Fit for purpose and proportionate to an internal business-hours finance system (no gold-plating)	Improvement Requirements
IR-3	Brought into compliance with the applicable Indian regulatory requirements	Indian Regulatory Requirements
IR-4	Application and financial data preserved — infrastructure only, no app change, no data loss	Improvement Requirements
IR-5	Operable and maintainable by YAT in-house ICT; monitoring/alerting; favour reproducible IaC	Improvement Requirements
IR-6	Each improvement justified on cost-versus-benefit; operating-cost impact made explicit	Improvement Requirements
IR-7	Production change follows Change Management; avoids the monthly close / EOFY Restricted Period	Improvement Requirements

3. Review of the Existing Architecture

The baseline is a single-AZ deployment in ap-southeast-2 (Sydney): an EC2 application tier (Amazon Linux, Auto Scaling group min 1 / max 2) in one Availability Zone behind an internal Application Load Balancer; an Amazon RDS for PostgreSQL database (single-AZ); document attachments on Amazon S3; gp3 EBS volumes; reached over the Site-to-Site VPN. It is functional but was provisioned for the migration, with high-availability hardening explicitly deferred.

3.1 Single points of failure

- The application tier runs in a single Availability Zone — an AZ or instance failure takes the system down.
- The database is a single instance with no standby, so an AZ or instance failure means an outage and a restore-from-backup. Ledgerline runs on a managed Multi-AZ database, so this single point of failure can be removed outright by the reliability improvements below.
- Network and data subnets exist in one AZ only, so nothing downstream can be made resilient until the network spans two AZs.

3.2 Current state against the four concerns and compliance

Concern	Current state	Gap
Security	Internal-only (VPN), encrypted at rest, security groups in place; patching/role posture not hardened	Tighten least-privilege, managed patching, secrets and TLS (encryption at rest is already in the baseline)
Reliability	Single-AZ compute and single-instance database; 99.5% business-hours target	Make the application tier Multi-AZ and convert the database to a Multi-AZ deployment with an automatic-failover standby, so both tiers survive an AZ/instance failure automatically; keep backup, point-in-time restore and cross-Region DR behind them
Scalability	ASG target-tracking exists; subnets/endpoints not sized for elastic growth	Confirm elastic-on-demand headroom across the tiers (not extra capacity for load not expected)
Cost	Instances run 24/7 though the system is idle overnight; storage untiered; SQL licensing not reviewed	Business-hours scheduling, storage lifecycle, right-sizing and a licensing review
Compliance	All logs and records sit in Sydney; no India residency	CERT-In logs (180-day, in India) and Companies-Act books retrievable in India

3.3 Goals and metrics

Concern	Goal	Metric
Security	Least-privilege, secrets-managed, managed-patch posture (baseline already encrypted at rest)	0 plaintext secrets; managed-patch compliance reported; least-privilege roles/SGs in place
Reliability	Application tier and database both survive an AZ/instance failure automatically; backup, point-in-time restore and cross-Region DR cover data loss, corruption and Region-level events	App tier: auto-recovery < 15 min on AZ/instance failure. Database: automatic failover under two minutes; RPO <= 1 hour (frequent log backups); cross-Region restore within the agreed business day

Scalability	Elastic on demand for the month-end/EOFY peak	Demonstrated scale-out under a controlled load test; no manual capacity change
Cost	Cut spend on the idle profile without losing service	Compute hours reduced outside business hours; storage cost/GB reduced via tiering
Compliance	India log + books residency met	CERT-In logs retained 180 days in ap-south-1; books-of-account retrievable in India

3.4 Review findings summary

Reviewed against IR-1 to IR-7, the improvement is: make the application tier Multi-AZ on a two-AZ network and convert the database to a Multi-AZ deployment with an automatic-failover standby (with backup, point-in-time restore and cross-Region DR behind it), harden security, realise the idle-profile and storage cost savings, confirm elastic headroom, and add the light India residency slice for regulatory logs and financial records. The application, its PostgreSQL stack and its data are preserved (IR-4); every change is proportionate and cost-justified (IR-2, IR-6).

4. Architecture Design

4.1 Assumptions and constraints

- The baseline is implemented and running; this design improves it in place, preserving the application and data (IR-4).
- The system stays internal (Site-to-Site VPN); there is no public ingress.
- Provisioning is parameterised CloudFormation across four component stacks (network / compute / database / storage), each independently authorable and deployable, integrated by cross-stack references.
- Production change follows Change Management and avoids the Restricted Period (monthly close, EOFY) (IR-7).

4.2 Accounts and regions

ap-southeast-2 (Sydney) remains the primary region for the whole system and its data — the DPDP Act permits the India-operation personal data to remain in Australia. ap-south-1 (Mumbai) is introduced for one purpose only: the light residency slice (§4.7) — the CERT-In system/access logs (retained in India for 180 days) and a periodic export of the India-operation books-of-account for Companies-Act retrievability. The main system and database stay in Sydney.

4.3 Design approach — four components

The improvement is expressed as four cloud-infrastructure components, each improved across all four optimisation concerns. They are provisioned as separate, parameterised CloudFormation stacks so they can be authored independently and integrated into one deployable system.

4.4 Component 1 — Network (VPC)

Concern	Improvement
Security	Tiered private subnets (app / data), NACLs between tiers, least-privilege security groups, VPC Flow Logs enabled (feed the India log slice); no public ingress, Site-to-Site VPN retained
Reliability	Extend the VPC across ap-southeast-2a and 2b — app and data subnets in both AZs, redundant egress — so the tiers above can be made AZ-resilient
Scalability	Subnet CIDRs sized with headroom; interface/gateway VPC endpoints (S3, SSM, Secrets)

	Manager, CloudWatch) so AWS-service traffic scales without a NAT bottleneck
Cost	VPC endpoints cut NAT-gateway and data-transfer charges; NAT sized to the business-hours profile

4.5 Component 2 — Compute (EC2 application tier)

Concern	Improvement
Security	Hardened Windows AMI, SSM Patch Manager for managed patching, least-privilege IAM instance role (no long-lived keys), security groups limited to ALB-to-app and app-to-database
Reliability	Auto Scaling group spanning 2a/2b behind the internal ALB; health-checked instances replaced automatically; an AZ loss leaves healthy capacity in the other
Scalability	Target-tracking on CPU (retained, absorbs the month-end peak); right-sized instance family; demonstrable scale-out under a controlled load test
Cost	Scheduled stop outside business hours (the system is idle overnight and at weekends — the largest single saving), right-sized instances, gp3 volumes

4.6 Component 3 — Database (RDS for PostgreSQL)

Concern	Improvement
Security	Encryption in transit (TLS); credentials in Secrets Manager with rotation; security group restricting access to the app tier; not publicly accessible (encryption at rest is already enabled in the baseline)
Reliability	Converted to a Multi-AZ deployment — a synchronously replicated standby in the second Availability Zone with automatic failover under two minutes, removing the database as a single point of failure. The application reaches the database by endpoint name, so failover needs no reconfiguration. Behind it: automated backups with point-in-time recovery, frequent transaction-log backups (RPO <= 1 hour), an automated cross-Region backup copy to a second Australian Region (Melbourne, ap-southeast-4) for disaster recovery — keeping financial data in Australia — and a tested restore runbook
Scalability	Right-sized instance class; storage autoscaling enabled; a read replica noted as headroom for month-end reporting (not provisioned by default — proportionate)
Cost	Right-sized to measured load; a Savings Plan / Reserved Instance for the steady baseline; no proprietary database licence to optimise — the managed PostgreSQL engine is a per-hour platform cost, unlike the per-core licensing the on-premises deployment carried; backup retention tuned (long-term financial-records retention handled by export to S3, not live RDS)

Reliability decision — cost versus benefit (IR-2, IR-6). Converting the database to Multi-AZ roughly doubles its instance and storage cost, because a standby of the same class runs alongside the primary. Weighed against the actual need — an internal, business-hours finance system with outsourced payroll, an estimated \$400/hour business-hours cost of downtime, and a two-hour recovery-time objective that a restore-only recovery does not reliably meet — the standby is proportionate and is recommended: it takes database recovery from hours to under two minutes, with no application change and no manual intervention. Backup, point-in-time restore and the cross-Region DR copy are retained behind it, because failover addresses AZ and instance failure but not data loss, corruption or a Region-level event. The application tier is made Multi-AZ on the same reasoning, at lower cost again.

4.7 Component 4 — Storage (S3 + EBS) and the India residency slice

Concern	Improvement
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Security	S3 Block Public Access; bucket policies; S3 access logging (encryption at rest is already enabled in the baseline)
Reliability	S3 durability with versioning for document attachments; automated EBS snapshots; lifecycle policies
Scalability	S3 absorbs unbounded document growth (~4 GB/yr) with no capacity planning; EBS gp3 volumes resized live (elastic volumes) as the footprint grows
Cost	S3 lifecycle / Intelligent-Tiering moves older scanned documents to IA/Glacier — a large saving over the 7-year retention

Residency slice (owned by this component): an ap-south-1 (Mumbai) S3 bucket holds (a) the CERT-In system/access logs — VPC Flow Logs, CloudTrail, ALB/application logs — with a ≥ 180 -day lifecycle and quick retrievability for the 6-hour reporting obligation, and (b) a periodic, encrypted, version-locked export of the India-operation books-of-account for Companies-Act retrievability. The main system and primary data remain in Sydney.

4.8 Monitoring and operability

CloudWatch metrics, dashboards and alarms per component provide the measures the goals (§3.3) are tested against and keep the environment operable by YAT in-house ICT (IR-5); CloudTrail records the audit trail (also feeding the India log slice). Alerting covers application-tier AZ failover, scaling events, and database backup / restore and lifecycle health.

4.9 Provisioning — infrastructure as code

The whole design is provisioned as parameterised CloudFormation, partitioned into four component stacks (network / compute / database / storage) with exported cross-stack references for the dependencies, so each can be authored, deployed and updated independently and the system reproduced consistently (IR-5).

4.10 Goals and metrics — designed state

Each goal in §3.3 is met by the design above: reliability by a Multi-AZ application tier and a Multi-AZ database with an automatic-failover standby on a two-AZ network, backed by database backup, point-in-time restore and cross-Region DR; scalability by the retained ASG target-tracking plus elastic storage and endpoint headroom (demonstrable on demand, not over-provisioned); cost by the business-hours schedule, storage tiering, right-sizing and the licensing review; security by the hardening, encryption and least-privilege posture; compliance by the India residency slice.

4.11 Single points of failure removed

Both single points of failure identified in §3.1 are removed. The application tier runs across two AZs behind the internal ALB on a network that now spans both AZs. The database runs as a Multi-AZ deployment with a synchronous standby in the second AZ and automatic failover, so an AZ or instance failure no longer takes the system down. Automated backups, point-in-time restore and the cross-Region DR copy remain in place to cover what failover does not: data loss, corruption and Region-level events (§4.6).

5. Implementation Sequencing

Changes are applied in dependency order, under Change Management and outside the Restricted Period (IR-7). The build is provisioned from the component CloudFormation stacks.

#	Change	Expected impact
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1	Network stack — extend the VPC to two AZs, add subnets, endpoints, flow logs	None — additive
2	Storage stack — encryption, versioning, lifecycle, and the ap-south-1 residency bucket	None — additive
3	Database stack — strengthen backups (frequent log backups, cross-Region copy to Melbourne), DR runbook; encryption in transit; secrets	None — additive; no failover reconfiguration (single instance retained)
4	Compute stack — ASG across two AZs, hardening, business-hours schedule	Rolling; capacity maintained throughout

6. Verification Plan

How the design is shown to meet the goals (§3.3); the evidence is produced at deployment.

- Reliability — terminate an app instance and confirm the ASG replaces it across AZs with no outage (application-tier auto-recovery); force a database failover and confirm the standby is promoted automatically, the application reconnects by endpoint name without reconfiguration, and service resumes inside two minutes; and perform a point-in-time restore and a cross-Region (Melbourne) DR restore, confirming recovery with RPO $\leq$ 1 hour and no data loss.
- Scalability — a controlled load test ramps to the month-end peak; confirm the ASG scales out and back and storage/endpoints absorb growth, with no manual capacity change.
- Security — confirm encryption at rest/in transit, least-privilege roles/SGs, patch compliance, and no plaintext secrets.
- Cost — confirm the business-hours schedule stops/starts compute and the storage lifecycle transitions older objects; report the operating-cost delta.
- Compliance — confirm CERT-In logs are written to and retained in ap-south-1 ($\geq$ 180 days) and the books-of-account export is retrievable in India.

7. Out of Scope

- The Ledgerline application and its financial data — preserved unchanged (IR-4).
- The legal interpretation of the Indian obligations — owned by YAT Compliance; treated here as fixed inputs.
- Payroll — outsourced to an external bureau; not on this system.
- Full localisation of the system into India — deliberately avoided; only the regulatory logs and books-of-account export sit in India (the light slice).

8. References

- Improvement Requirements (Ledgerline Cloud Infrastructure Improvement)
- Indian Regulatory Requirements (DPDP Act 2023; CERT-In Directions 2022; Companies Act 2013 / Income-tax retention)
- Accounting System Cloud Architecture — Baseline Design
- Accounting System Infrastructure Specifications; Operational Costing
- AWS Well-Architected Framework — Reliability, Security, Cost Optimization and Performance Efficiency pillars

9. Review and Approval

The design was reviewed against IR-1 to IR-7 and the goals (§3.3). It is presented to the required personnel for review, and sign-off to proceed to deployment is obtained at that presentation; the record below captures that approval.

Role	Name	Decision	Date
Prepared by	MTS Improvement Team	Submitted	[date]
Reviewed by	MTS Senior Consultant	Approved for submission	[date]
Approved by	Sam Walker (YAT ICT Manager)	Approved — cleared to proceed to deployment	[date]

Document control

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